

Absolute Value Functions

Name _____

Date _____

Part A — Vertex & Graphing. State the vertex (h, k) , say whether the V opens *up* or *down*, and sketch the graph on your own grid.

- 1) $f(x) = |x - 2| + 1$
- 2) $f(x) = -|x + 3|$
- 3) $f(x) = 2|x| - 4$
- 4) $f(x) = -\frac{1}{2}|x - 1| + 3$
- 5) $f(x) = |x + 4| - 2$

Part B — Writing Equations. Write an absolute value function $f(x) = a|x - h| + k$ matching each description.

- 6) Vertex $(3, -2)$, opens up, same width as the parent $|x|$.
- 7) Vertex $(-1, 5)$, opens down, branch slopes ± 3 .
- 8) Vertex at the origin, vertically compressed by a factor of $\frac{1}{2}$.

Part C — Solving Equations. Solve each equation. Write “no solution” where appropriate.

- 9) $|x - 4| = 9$
- 10) $|2x + 1| = 7$
- 11) $3|x| - 5 = 10$
- 12) $|x + 6| + 8 = 3$
- 13) $2|3x - 1| + 5 = 15$
- 14) $|5 - x| = 0$
- 15) $\frac{1}{4}|x + 2| = 3$

Part D — Solving Inequalities. Solve each inequality. Write the solution in *interval notation* and graph it on a number line.

- 16) $|x| < 6$
- 17) $|x - 3| \leq 5$
- 18) $|2x + 1| > 7$
- 19) $|x + 4| \geq 2$
- 20) $|3x - 2| < 10$
- 21) $2|x - 1| + 3 > 9$

Part E — Reasoning.

- 22) Explain why $|x - 5| = -2$ has no solution.
- 23) The graph of $f(x) = a|x - h| + k$ has vertex $(2, -3)$ and passes through $(4, 1)$. Find a .
- 24) Describe, in order, how the graph of $f(x) = -3|x + 1| + 4$ is transformed from the parent $|x|$ (use $D \rightarrow R \rightarrow S$).